Vector Analysis and Geometry

2MBC40

The source code for these notes is publicly available in my github.
Contributions and corrections are always welcome.

`https://github.com/Luiser13/vector-analysis-summary`

Luis Rodriguez Pacheco

Eindhoven University of Technology

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Useful Tools Not Covered in the Course

Coordinate transformations

Polar

$$x = r\cos\theta, \quad y = r\sin\theta$$
$$r \in [0,\infty),\ \theta \in [0,2\pi]$$
$$\det(D\Phi_{\text{pol}}) = r, \quad dA = r\,dr\,d\theta$$

Elliptic Polar

$$x = x_0 + ar\cos\theta, \quad y = y_0 + br\sin\theta$$
$$r \in [0,1],\ \theta \in [0,2\pi]$$
$$\det(D\Phi_{\text{ell}}) = abr, \quad dA = abr\,dr\,d\theta$$

Cylindrical

$$x = r\cos\theta, \quad y = r\sin\theta, \quad z = z$$
$$r \in [0,\infty),\ \theta \in [0,2\pi],\ z \in \mathbb{R}$$
$$\det(D\Phi_{\text{cyl}}) = r, \quad dV = r\,dr\,d\theta\,dz$$

Spherical

$$x = \rho\sin\phi\cos\theta$$
$$y = \rho\sin\phi\sin\theta$$
$$z = \rho\cos\phi$$
$$\rho \in [0,\infty),\ \theta \in [0,2\pi],\ \phi \in [0,\pi]$$
$$\det(D\Phi_{\text{sph}}) = \rho^2\sin\phi,$$
$$dV = \rho^2\sin\phi\,d\rho\,d\phi\,d\theta$$

Weierstrass substitution

Useful for turning trigonometric integrals into rational integrals.

$$t = \tan\left(\frac{x}{2}\right), \qquad dx = \frac{2}{1+t^2}\,dt$$
$$\sin x = \frac{2t}{1+t^2}, \qquad \cos x = \frac{1-t^2}{1+t^2}$$

When to use:

- Integrands involving only $\sin x$, $\cos x$, and rational operations.
- Especially useful when standard identities do not simplify the integral enough.

Important: For a full period $x \in [0,2\pi]$, split at $x = \pi$, since

$$\tan\left(\frac{\pi}{2}\right) \text{ is undefined.}$$

Procedure:

1. Set $t = \tan(x/2)$.
2. Replace $\sin x, \cos x, dx$ using the formulas above.
3. Simplify to a rational function in t.
4. Integrate using algebraic methods: partial fractions, symmetry, completing the square.

Part I

Classical Vector Analysis

Chapter 1

Euclidean Space and Fields

1.1 Euclidean Space $\mathbb{R}^3$

Definition 1.1 (Inner Product)**.**

$$\left(\overrightarrow{OX}, \overrightarrow{OY}\right) = |\overrightarrow{OX}|\,|\overrightarrow{OY}|\cos\big(\angle(\overrightarrow{OX}, \overrightarrow{OY})\big).$$

Definition 1.2 (Cross Product)**.**

$$\overrightarrow{OX} \times \overrightarrow{OY} = \overrightarrow{OZ}$$

such that:

1. $\overrightarrow{OX} \perp \overrightarrow{OZ}$ and $\overrightarrow{OY} \perp \overrightarrow{OZ}$, that is,

$$\left(\overrightarrow{OX}, \overrightarrow{OZ}\right) = \left(\overrightarrow{OY}, \overrightarrow{OZ}\right) = 0.$$

2. $$|\overrightarrow{OZ}| = |\overrightarrow{OX}|\,|\overrightarrow{OY}|\sin(\alpha),$$

   so $|\overrightarrow{OZ}|$ is the area of the parallelogram spanned by $\overrightarrow{OX}$ and $\overrightarrow{OY}$.

3. If $|\overrightarrow{OZ}| \neq 0$, then the ordered set (OX, OY, OZ) is right-handed.

1.2 Scalar Fields

Definition 1.3. A scalar field φ defined on $D \subset \mathbb{R}^3$ is a map

$$\varphi : D \subset \mathbb{R}^3 \to \mathbb{R}.$$

1.3 Vector Fields

A vector field F defined on $D \subset \mathbb{R}^3$ associates to each point $P \in D$ a vector $F(P)$. With basis $\{e_1(P), e_2(P), e_3(P)\}$, the field represents the vector F at P as

$$F(P) = F_1(P)e_1(P) + F_2(P)e_2(P) + F_3(P)e_3(P).$$

In $\mathbb{R}^3$, a vector field is a map $F : \mathbb{R}^3 \to \mathbb{R}^3$ such that for each point $(x, y, z) \in \mathbb{R}^3$,

$$F(x, y, z) = \big(F_1(x, y, z), F_2(x, y, z), F_3(x, y, z)\big) \in \mathbb{R}^3.$$

1.4 Field Lines

Definition 1.4. Let F be a vector field on D. The field lines of F are smooth curves K such that for every $p \in K$, the vector $F(p)$ is tangent to K.

The parametrized curve

$$r(t) = x(t)i + y(t)j + z(t)k$$

is a field line for the vector field F if

$$\dot{r}(t) = \frac{dr(t)}{dt} = \lambda(t)\, F(r(t)), \qquad \lambda(t) > 0.$$

In polar coordinates,

$$F(r,\theta) = F_r(r,\theta)\,\hat{r} + F_\theta(r,\theta)\,\hat{\theta},$$

with field line given by the system

$$\frac{dr(t)}{dt} = \lambda(t)\, F_r\big(r(t),\theta(t)\big)$$

and

$$r(t)\frac{d\theta(t)}{dt} = \lambda(t)\, F_\theta\big(r(t),\theta(t)\big).$$

1.5 Potentials and Conservative Vector Fields

If

$$F(x,y,z) = \nabla\varphi(x,y,z) \equiv \left(\frac{\partial\varphi}{\partial x}, \frac{\partial\varphi}{\partial y}, \frac{\partial\varphi}{\partial z}\right)$$

on a domain D, then F is conservative on D. Also, the scalar field φ is the potential of F on D.

Chapter 2

Curves, Line Integrals, and Surfaces

2.1 Curves and Arc Length

Definition 2.1. A curve C from a to b is the set of points traced out by a smooth path $r : [\alpha, \beta] \to \mathbb{R}^n$ such that:

1. $r \in C^k$,
2. $r(\alpha) = a$ and $r(\beta) = b$,
3. r is a bijection,
4. $r'(t) \neq 0$ for all $t \in (\alpha, \beta)$.

Then r is a parametrization of the curve C.

Remark 2.2. k is generally ∞.

Definition 2.3 (Piecewise Smoothness)**.** A piecewise smooth curve is a path made of a finite number of smooth segments.

Note 2.4. A closed curve is a curve with parametrization $r : [a, b] \to \mathbb{R}^n$ such that $r(a) = r(b)$, so it is not injective on the whole interval.

Let P be a partition of $[a, b]$ and let $r(t)$ be a parametrization. We take

$$P: \ a = t_0 < t_1 < \cdots < t_N = b.$$

Then

$$L_P = \sum_{k=0}^{N-1} |r(t_{k+1}) - r(t_k)|,$$

and the arc length of a smooth curve C is given by

$$L(C) = \sup\{L_P \mid P \text{ is a partition of } [a, b]\} = \lim_{|P| \to 0} L_P.$$

Theorem 2.5. *Let C be a smooth curve with smooth parametrization*

$$r(t) = \big(r_1(t), \ldots, r_n(t)\big), \qquad t \in [a, b].$$

Then

$$L(C) = \int_a^b \sqrt{r_1'(t)^2 + r_2'(t)^2 + \cdots + r_n'(t)^2}\, dt = \int_a^b |r'(t)|\, dt.$$

Let $r : [a, b] \to \mathbb{R}^n$ be a smooth parametrization of C and let

$$a = t_0 < \cdots < t_N = b$$

be a partition. Approximating C by segments yields

$$\sum_{i=0}^{N-1} f(r(t_i))\,|r(t_{i+1}) - r(t_i)|.$$

By the Mean Value Theorem,

$$|r(t_{i+1}) - r(t_i)| = |r'(\tau_i)|(t_{i+1} - t_i), \qquad \tau_i \in (t_i, t_{i+1}).$$

Hence the sum becomes

$$\sum f(r(t_i))|r'(\tau_i)|\,\Delta t_i,$$

so

$$\lim_{\Delta t_i \to 0} \sum f(r(t_i))|r'(\tau_i)|\,\Delta t_i$$

is a Riemann sum that converges to

$$\int_a^b f(r(t))|r'(t)|\,dt.$$

Theorem 2.6. *The line integral along the curve C is*

$$\int_C f\,ds = \int_a^b f(r(t))|r'(t)|\,dt.$$

2.2 Work, Circulation, and Path Independence

2.2.1 Line Integral of a Vector Field (Work)

Let C be a piecewise smooth curve, let F be a continuous vector field, and let

$$\mathbf{t} := \frac{r'(t)}{|r'(t)|}$$

be the unit tangent vector, where $r(t)$ is a parametrization of C. Then $(F, \mathbf{t})$ exists and

$$\int_C (F, \mathbf{t})\ ds$$

is the work done by F along C.

Note 2.7. Since $ds = |r'(t)|dt$ and $\mathbf{t} = \dfrac{r'(t)}{|r'(t)|}$, then

$$\int_C (F, \mathbf{t})\ ds = \int_a^b \big(F(r(t)), r'(t)\big)\ dt,$$

where $r(a)$ and $r(b)$ are the endpoints of C.

Definition 2.8. If $r(a) = r(b)$, then C is a closed curve. If intersections may be present, then we have the cyclic integral

$$\oint_C f\,ds = \oint_C F \cdot dr,$$

the circulation of F around C.

2.2.2 Path Independence

In a conservative vector field, the work done is independent of path.

Theorem 2.9. *Let F be conservative with potential φ. Then*

$$\int_C (F, \mathbf{t})\ ds = \int_C (\nabla\varphi, \mathbf{t})\ ds = \int_a^b \big(\nabla\varphi(r(t)), r'(t)\big)\ dt = \varphi(r(b)) - \varphi(r(a)).$$

Note 2.10. Then

$$\oint_C (F, \mathbf{t})\ ds = \varphi(r(b)) - \varphi(r(a)) = 0.$$

2.3 Surface Integrals

Note 2.11. $\dfrac{\partial x}{\partial u}$ and $\dfrac{\partial x}{\partial v}$ span the tangent plane.

Definition 2.12. A surface $S \subset \mathbb{R}^3$ is called smooth if there exist $G \subset \mathbb{R}^2$ and a map $x : G \to \mathbb{R}^3$, continuously differentiable and bijective, such that

$$\frac{\partial x}{\partial u} \times \frac{\partial x}{\partial v} \neq 0 \qquad \forall (u, v) \in G,$$

so $S = x(G)$.

Theorem 2.13. *Let f be a continuous scalar field. Then*

$$\iint_S f\, d\sigma = \iint_G f\big(x(u, v)\big) \left| \frac{\partial x}{\partial u} \times \frac{\partial x}{\partial v} \right| du\, dv.$$

Intuition 2.14. $d\sigma$ is the surface-area element on S. When we change variables to the flat domain G, we still want to integrate with respect to the surface area of S. Therefore the values of f must be taken at points on the surface, hence $f(x(u, v))$, and

$$\left| \frac{\partial x}{\partial u} \times \frac{\partial x}{\partial v} \right|$$

accounts for how area in G scales to surface area on S.

2.4 Orientation

The unit normal $n(P)$ at the point $P = x(u_p, v_p)$ is

$$n(P) := \pm \frac{\dfrac{\partial x}{\partial u} \times \dfrac{\partial x}{\partial v}}{\left| \dfrac{\partial x}{\partial u} \times \dfrac{\partial x}{\partial v} \right|} (u_p, v_p).$$

The sign is determined by checking whether n and

$$\frac{\partial x}{\partial u} \times \frac{\partial x}{\partial v}$$

point in the same direction:

$$\left(n, \frac{\partial x}{\partial u} \times \frac{\partial x}{\partial v}\right) > 0 \iff \text{positive orientation},$$

and

$$\left(n, \frac{\partial x}{\partial u} \times \frac{\partial x}{\partial v}\right) < 0 \iff \text{negative orientation}.$$

2.5 Flux

Definition 2.15. The flux of F through S is

$$\iint_S (F, n)\ d\sigma.$$

It follows that

$$\iint_S (F, n)\ d\sigma = \pm \iint_G \left(F\big(x(u,v)\big), \frac{\partial x}{\partial u} \times \frac{\partial x}{\partial v}\right)\ du\, dv.$$

Chapter 3

Differential Operators and Integral Theorems

3.1 Gradient, Divergence, and Curl

Gradient:

$$\nabla\varphi = \frac{\partial\varphi}{\partial x}i + \frac{\partial\varphi}{\partial y}j + \frac{\partial\varphi}{\partial z}k.$$

Jacobian:

$$DF = \begin{pmatrix} \frac{\partial F_1}{\partial x_1} & \cdots & \frac{\partial F_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial F_n}{\partial x_1} & \cdots & \frac{\partial F_n}{\partial x_n} \end{pmatrix}.$$

Divergence:

$$\operatorname{div} F = \operatorname{trace}(DF).$$

If $\operatorname{div} F = 0$, then F is solenoidal.

Curl:

$$\operatorname{curl} F = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\right) i + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}\right) j + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) k.$$

Note that $\operatorname{curl} F = \nabla \times F$.

Theorem 3.1. *If a vector field has* $\operatorname{curl} F = 0$*, it is irrotational.*

3.2 Laplacian Operator of a Scalar Field

$$\Delta := \nabla \cdot \nabla, \qquad \Delta\varphi = \operatorname{div}(\nabla\varphi).$$

The scalar field φ is harmonic on D if $\Delta\varphi = 0$.

3.3 Identities

$$\nabla(\varphi\psi) = \varphi\nabla\psi + \psi\nabla\varphi \qquad \nabla \times (\nabla\varphi) = 0$$

$$\operatorname{div}(\varphi F) = \varphi \operatorname{div} F + (\nabla\varphi) \cdot F \qquad \operatorname{div}(\operatorname{curl} F) = 0$$

$$\nabla \times (\varphi F) = \varphi(\nabla \times F) + \nabla\varphi \times F$$

$$\operatorname{div}(F \times G) = (G, \nabla \times F) - (F, \nabla \times G) \qquad \text{with } (F \cdot \nabla)G := DG[F]$$

$$\nabla \times (\nabla \times F) = \nabla(\nabla \cdot F) - (\nabla \cdot \nabla)F$$

$$\nabla \times (F \times G) = F(\nabla \cdot G) - G(\nabla \cdot F) + (G \cdot \nabla)F - (F \cdot \nabla)G$$

$$\nabla(F \cdot G) = F \times (\nabla \times G) + G \times (\nabla \times F) + (G \cdot \nabla)F + (F \cdot \nabla)G$$

3.4 Integral Theorems

Recall that

$$\mathbf{t} = \frac{r'(t)}{|r'(t)|} \qquad \text{and} \qquad ds = |r'(t)|\, dt.$$

3.4.1 Gauss' Theorem (Divergence Theorem)

Let $R \subset \mathbb{R}^3$ be a region with piecewise smooth boundary ∂R, and let n be the outward unit normal. Let F and φ be continuously differentiable on a domain containing R. Then

$$\iiint_R \nabla\varphi \, dV = \oiint_{\partial R} \varphi n \, d\sigma,$$

which is the vector form.

Also,

$$\iiint_R \operatorname{div}(F) \, dV = \oiint_{\partial R} (F, n) \; d\sigma.$$

3.4.2 Stokes' Theorem

For $\varphi, F \in C^1(D)$, it holds that

$$\iint_S n \times \nabla\varphi \, d\sigma = \oint_{\partial S} \varphi \mathbf{t} \, ds$$

and

$$\iint_S (\operatorname{curl} F, n) \; d\sigma = \oint_{\partial S} (F, \mathbf{t}) \; ds.$$

3.4.3 Green's Theorem

This is a special case of Stokes' theorem for $n = (0, 0, 1)$.

Let $D \subset \mathbb{R}^2$ be open and connected, and let $R \subset D$ be a domain with piecewise smooth positively oriented boundary ∂R. Let $F : D \to \mathbb{R}^2$. Then

$$\iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dA = \oint_{\partial R} F_1(x, y)\, dx + F_2(x, y)\, dy = \oint_{\partial R} (F, \mathbf{t})\ ds.$$

3.5 Vector Potential

A vector field F has a vector potential if

$$F = \operatorname{curl} G.$$

A continuous vector field F is source free if

$$\oiint_S (F, n)\ d\sigma = 0,$$

so there is zero flux for closed S in D.

If F is solenoidal, that is, if $\operatorname{div}(F) = 0$, then F is source free.

Theorem 3.2 (Helmholtz Decomposition)**.** *For any $F : \mathbb{R}^3 \to \mathbb{R}^3$ of class C^2, we can decompose it into*

$$-\nabla\phi + \nabla \times A,$$

that is, an irrotational part plus a solenoidal part.

Note 3.3. If $\operatorname{div} F = 0$ then

$$G := \int_0^1 t(F(tx) \times x)\, dt$$

is a vector potential.

To fulfill hypothesis $\mathcal{H}$, a subset $D \subset \mathbb{R}^3$ needs:

- For any closed surface $S \subset D$, the domain enclosed by S is completely contained in D.
- D is non-periphractic, so there are no holes in D.

Part II

Modern Vector Analysis

Chapter 4

Manifolds

4.1 Topological Manifolds

A topological manifold is a space M such that for every $p \in M$, there exists an open neighborhood $U \subset M$ containing p and a homeomorphism

$$\varphi : U \to V \subset \mathbb{R}^n,$$

where V is open.

A space that globally is weird but locally Euclidean.

The pair (U, φ) is a chart, and it is centered if there exists $p \in U$ such that $\varphi(p) = 0$.

4.2 Smooth Atlases

This is a collection

$$\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}$$

with $\bigcup_\alpha U_\alpha = M$, where M is an n-dimensional manifold. Here, any two charts in $\mathcal{A}$ are smoothly compatible.

Two charts are smoothly compatible if $U_\alpha \cap U_\beta = \varnothing$ or if

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \ \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$$

and

$$\varphi_\alpha \circ \varphi_\beta^{-1} : \ \varphi_\beta(U_\alpha \cap U_\beta) \to \varphi_\alpha(U_\alpha \cap U_\beta)$$

are smooth.

Note 4.1. Smoothly compatible is also C^∞-compatible.

4.3 Maximal Atlases

A maximal atlas $\mathcal{A}$ contains all charts on M that are smoothly compatible.

4.4 Smooth Manifolds

A smooth manifold is a pair $(M, \mathcal{A})$ having a differential structure defined by $\mathcal{A}$.

A scalar function $f : M \to \mathbb{R}$ is smooth at $p \in M$ if in a chart,

$$f \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}$$

is smooth at $\varphi(p)$.

Intuition 4.2. A differential structure is the maximal collection of charts that are smoothly compatible on M and that allows us to build a local pipeline

$$\mathbb{R}^n \xrightarrow{\varphi^{-1}} U \subset M \xrightarrow{f} \mathbb{R}$$

so that we can perform calculus independently of chart choice.

4.5 Oriented Manifolds

A smooth manifold M is oriented if its maximal atlas

$$\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}$$

is such that the Jacobian determinant of each transition map $\varphi_\beta \circ \varphi_\alpha^{-1}$ is positive on overlaps.

4.6 Submanifolds

A subset $K \subset M$ is called a submanifold of dimension $k \leq m$ if, for every point $p \in K$, there exists a chart (U, φ) with $p \in U \subset M$ such that

$$\varphi(U \cap K) = \varphi(U) \cap \iota(\mathbb{R}^k) = \{(x^1, \ldots, x^m) \in \varphi(U) \subset \mathbb{R}^m \mid x^{k+1} = x^{k+2} = \cdots = x^m = 0\},$$

where $\iota : \mathbb{R}^k \to \mathbb{R}^m$ is the inclusion map.

Note 4.3. A submanifold is a manifold.

4.7 Manifolds with Boundary

An n-dimensional manifold M with boundary is a space such that every point $p \in M$ admits a chart into either $\mathbb{R}^n$ or the half-space H^n, with

$$H^n = \{(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n \mid x_n \geq 0\}.$$

Transition maps between these charts are smooth.

Define the boundary by

$$\partial M := \{p \in M : \varphi(p) \in \{x_n = 0\}\}.$$

Define the interior by

$$M^\circ := \{p \in M : \varphi(p) \in \{x_n > 0\}\}.$$

Theorem 4.4. *∂M is a boundary-less $(n-1)$-manifold acting as the boundary of M.*

Example 4.5. Let $M := T^2 \times [0,1]$, where T^2 is a torus $S^1 \times S^1$.

The local coordinates for T^2 are (θ, ϕ) and t for $[0,1]$, therefore the coordinates for M are $(\theta, \phi, t) \in [0, 2\pi) \times [0, 2\pi) \times [0,1]$.

The boundary ∂M is then given by

$$(T^2 \times \{0\}) \cup (T^2 \times \{1\}).$$

The interior M° is therefore $T^2 \times (0,1)$.

Chapter 5

Tangent and Cotangent Structures

5.1 Tangent Space

This is the set of all possible derivations at p:

$$T_pM = \{v_p : C^\infty(M) \to \mathbb{R} \mid v_p \text{ is a derivation at } p\}.$$

Here v_p satisfies the linearity and Leibniz rules:

- $v_p[\alpha f + \beta g] = \alpha v_p[f] + \beta v_p[g]$,
- $v_p[fg] = f(p)\, v_p[g] + g(p)\, v_p[f]$.

In local coordinates $(x^1, \ldots, x^n)$, every tangent vector can be written as

$$v_p = v^i \frac{\partial}{\partial x^i}\Big|_p.$$

A basis for T_pM with local coordinates $(x^1, x^2, x^3, \ldots, x^n)$ is

$$T_pM = \operatorname{span}\left\{ \frac{\partial}{\partial x^1}\Big|_p, \ldots, \frac{\partial}{\partial x^n}\Big|_p \right\}.$$

Note 5.1. These are not powers; they are indices.

5.2 Tangent Bundle

Defined as

$$TM = \bigsqcup_{p \in M} T_pM.$$

All elements in TM have the form (p, v_p), where $p \in M$ and $v_p \in T_pM$.

Projection: map $\pi : TM \to M$, $\pi(p, v_p) = p$.

Fiber: $\pi^{-1}(p) = T_pM$.

Theorem 5.2. *If M has dimension n, then TM has dimension $2n$.*

Definition 5.3. A vector field on M is a smooth section of TM that assigns to each point of M a vector in the fiber $\pi^{-1}(p) = T_pM$.

Note 5.4. $\mathfrak{X}(M)$ denotes the space of all vector fields on M.

5.3 Dual Spaces

The dual space V^* of V is the set of linear maps $V \to \mathbb{R}$ that measure vectors in V. They both have the same dimension.

If V has basis $\{e_1, e_2, \ldots, e_n\}$, then we can define a dual basis $\{\mu^1, \mu^2, \ldots, \mu^n\}$ of V^* by imposing that

$$\mu^i(e_j) = \delta^i_j.$$

Elements in V^* are covectors, and applying $\alpha \in V^*$ to $v \in V$ is the canonical pairing:

$$\langle \alpha, v \rangle := \alpha(v).$$

$$\mu^i(v) = \langle \mu^i, v \rangle = \left\langle \mu^i, v^j e_j \right\rangle = v^j \langle \mu^i, e_j \rangle = v^j \delta^i_j = v^i.$$

$$\delta^i_j = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

5.4 Cotangent Space

$T^*_p M$ is the dual space of $T_p M$, where its elements are covectors. It is spanned by

$$\left\{ dx^1, dx^2, \ldots, dx^n \right\}.$$

$$T^* M = \bigsqcup_{p \in M} T^*_p M,$$

exactly as in the tangent bundle, with elements (p, ξ) where $p \in M$ and $\xi \in T^*_p M$. The projection is π^* and the fiber is $(\pi^*)^{-1}(p)$.

5.5 Differential 1-Forms

A differential 1-form is a smooth section of the cotangent bundle that assigns to each point p a covector $\omega_p \in T^*_p M$.

Equivalently, it acts on vector fields as

$$\omega : \mathfrak{X}(M) \to C^\infty(M).$$

Thus it assigns a linear map that measures tangent vectors at each point, from the fiber $(\pi^*)^{-1}(p) \subset T^* M$.

Note 5.5. $\Omega^1(M)$ denotes the space of all differential 1-forms.

Chapter 6

Tensors, Metrics, and Differential Forms

6.1 Tensors

A tensor of type (q, r) is a multilinear function defined as follows: for $p \in M$,

$$\mathcal{T}_p : \underbrace{T_p^*M \times \cdots \times T_p^*M}_{q} \times \underbrace{T_pM \times \cdots \times T_pM}_{r} \longrightarrow \mathbb{R}.$$

So we have

$$\mathcal{T}_p(\alpha^1, \ldots, \alpha^q, v_1, \ldots, v_r) \equiv \sum_{i_1=1}^{n} \cdots \sum_{i_q=1}^{n} \sum_{j_1=1}^{n} \cdots \sum_{j_r=1}^{n} \mathcal{T}^{i_1,\ldots,i_q}_{j_1,\ldots,j_r}\, \alpha^1_{i_1} \cdots \alpha^q_{i_q}\, v_1^{j_1} \cdots v_r^{j_r}.$$

The components of $\mathcal{T}_p$ are defined as

$$\mathcal{T}^{i_1,\ldots,i_q}_{j_1,\ldots,j_r} = \mathcal{T}_p\left(dx^{i_1}, \ldots, dx^{i_q}, \frac{\partial}{\partial x^{j_1}}, \ldots, \frac{\partial}{\partial x^{j_r}}\right).$$

6.2 Riemannian Manifolds

A Riemannian metric g is a smooth assignment of g_p, an inner product (a tensor of type $(0, 2)$), on T_pM. So we have

$$g_p : T_pM \times T_pM \to \mathbb{R}.$$

Note that g_p is linear, symmetric, and positive definite. A pair (M, g) is a Riemannian manifold.

6.3 Ricci Calculus

The flat operator $\flat : T_pM \to T_p^*M$ converts vectors to covectors. Note that

$$\langle v^\flat, w\rangle = g_p(v, w), \qquad \forall w \in T_pM.$$

The sharp operator $\sharp : T_p^*M \to T_pM$ converts covectors to vectors. Note that

$$g_p(\omega^\sharp, v) = \langle \omega, v\rangle, \qquad \forall v \in T_pM.$$

In a coordinate basis,

$$g_p(v, w) = g_p\left(v^i \frac{\partial}{\partial x^i}, w^j \frac{\partial}{\partial x^j}\right) = g_{ij} v^i w^j,$$

with g_{ij} the (i,j)-entry of the Gram matrix $G = (g_{ij})$ with inverse $G^{-1} = (g^{ij})$, such that

$$g^{ik} g_{kj} = \delta^i_j.$$

Note that

$$(v^\flat)_j = g_{ij} v^i \qquad \text{and} \qquad (\omega^\sharp)^i = g^{ij} \omega_j.$$

6.4 Differential k-Forms

A differential k-form ω on M is a smooth assignment of an alternating k-tensor ω_p acting on k vectors:

$$\omega_p : T_pM \times T_pM \times \cdots \times T_pM \to \mathbb{R}$$

such that $\omega_p(v_1, v_2, \ldots, v_k) = 0$ whenever $v_1, \ldots, v_k$ is a linearly dependent set in T_pM.

Theorem 6.1. *An alternating k-tensor ω_p is also skew-symmetric. So*

$$\omega_p\big(v_{\pi(1)}, v_{\pi(2)}, \ldots, v_{\pi(k)}\big) = \operatorname{sign}(\pi)\, \omega_p(v_1, \ldots, v_k).$$

Recall that $\operatorname{sign}(\pi) = 1$ *if* π *is even and* -1 *if* π *is odd.*

So a differential k-form is a smooth section of $\Lambda^k(T^*M)$, that assigns to each point p an alternating k-tensor. Here $\Lambda^k(T^*M)$ is the alternating k-tensor bundle.

For clarity, in coordinates:

1-form:

$$\omega = \sum_{i=1}^n \omega_i \, dx^i, \qquad \text{for example } \omega = F_1 dx^1 + F_2 dx^2 + F_3 dx^3 \text{ in } \mathbb{R}^3.$$

2-form:

$$\eta = \sum_{i<j} \eta_{ij} \, dx^i \wedge dx^j.$$

k-form:

$$\alpha = \sum_{i_1 < \cdots < i_k} \alpha_{i_1, \ldots, i_k} \, dx^{i_1} \wedge \cdots \wedge dx^{i_k}.$$

Note 6.2. There are $\binom{n}{k}$ linearly independent basis alternating k-tensors.

6.5 Exterior Algebra

Let $\mu \in \Omega^k(M)$ and $\nu \in \Omega^\ell(M)$. The wedge product is a $(k+\ell)$-differential form, so

$$\mu \wedge \nu \in \Omega^{k+\ell}(M).$$

In coordinates,

$$\mu \wedge \nu = \mu_{i_1, \ldots, i_k} \nu_{j_1, \ldots, j_\ell} \, dx^{i_1} \wedge \cdots \wedge dx^{i_k} \wedge dx^{j_1} \wedge \cdots \wedge dx^{j_\ell}.$$

The wedge product has properties:

- Distributive: $(\mu + \nu) \wedge \eta = \mu \wedge \eta + \nu \wedge \eta$.
- Associative: $(\mu \wedge \nu) \wedge \eta = \mu \wedge (\nu \wedge \eta)$.
- Anti-commutative: $\mu \wedge \eta = (-1)^{k\ell}(\eta \wedge \mu)$, where $\mu \in \Omega^k(M)$ and $\eta \in \Omega^\ell(M)$.
- Alternating: $dx^i \wedge dx^i = 0$ for all i.

Remark 6.3. $\mu_{i_1,\ldots,i_k}$ are the components of μ such that $i_1 < i_2 < \cdots < i_k$.

Note 6.4. For a 0-form α, $\wedge$ is the same as ordinary multiplication: $\alpha \wedge \mu = \alpha\mu$.

Example 6.5. In $\mathbb{R}^3$, let $\nu \in \Omega^2(M)$,

$$\nu_p = 2dx^1 \wedge dx^2 - 3dx^1 \wedge dx^3 + 5dx^2 \wedge dx^3 = \sum_{i<j} \nu_{ij}\, dx^i \wedge dx^j.$$

So then $\nu_{12} = 2$, $\nu_{13} = -3$, and $\nu_{23} = 5$.

Theorem 6.6. *The wedge product is an antisymmetrized tensor product. Formally,*

$$(\omega \wedge \mu)_p = \frac{(k+\ell)!}{k!\,\ell!}\, \mathcal{O}_a(\omega_p \otimes \mu_p),$$

where $\omega \in \Omega^k(M)$ and $\mu \in \Omega^\ell(M)$.

$\mathcal{O}_a$ is the skew-symmetrization map, for which a $\binom{k}{0}$-tensor $\mathcal{T}_p$ on T_pM is defined by

$$\mathcal{O}_a(\mathcal{T}_p)(v_1, \ldots, v_k) = \frac{1}{k!} \sum_\pi \operatorname{sgn}(\pi)\, \mathcal{T}_p(v_{\pi(1)}, \ldots, v_{\pi(k)}).$$

6.6 Exterior Derivative

The exterior derivative d is an operator that raises the degree of a differential form by 1:

$$d : \Omega^k(M) \to \Omega^{k+1}(M) : \omega \mapsto d\omega,$$

for all $k \leq n$, where $\dim(M) = n$.

Properties:

- Differential rule: if $\alpha \in \Omega^0(M)$, then $d\alpha$ is the differential, so

$$d\alpha = \sum_i \frac{\partial \alpha}{\partial x^i}\, dx^i.$$

- Nilpotency: $d(d\alpha) = 0$, hence $d^2 = 0$.
- Product rule: $d(\omega \wedge \mu) = d\omega \wedge \mu + (-1)^k \omega \wedge d\mu$.
- Distributivity: $d(\omega + \mu) = d\omega + d\mu$.

Also note that $d\omega$ is defined pointwise for all $p \in M$ by

$$(d\omega)_p = d(\omega_p).$$

6.7 Closed and Exact Forms

A differential form ω is closed if $d\omega = 0$.

A differential k-form is exact if there exists $\mu \in \Omega^{k-1}(M)$ such that $\omega = d\mu$.

Note that exact $\Rightarrow$ closed, but closed $\not\Rightarrow$ exact.

6.8 Contraction Operator

Definition 6.7. Let M be an n-dimensional manifold. With every vector field $v \in \mathfrak{X}(M)$ we can associate a linear map ι_v, a contraction. This ι_v maps a differential k-form to a $(k-1)$-form, so

$$\iota_v : \Omega^k(M) \to \Omega^{k-1}(M) : \omega \mapsto \iota_v \omega,$$

defined by

$$(\iota_v \omega)(v_1, \ldots, v_{k-1}) = \omega(v, v_1, \ldots, v_{k-1})$$

for all $v_1, v_2, \ldots, v_{k-1} \in \mathfrak{X}(M)$.

If $k = 0$, then the target would be $\Omega^{-1}(M)$, so we set $\iota_v f = 0$.

Properties:

Antisymmetry:

$$\iota_v \circ \iota_w = -\iota_w \circ \iota_v.$$

Product rule:

$$\iota_v(\mu \wedge \omega) = (\iota_v \mu) \wedge \omega + (-1)^\ell \mu \wedge (\iota_v \omega),$$

where $\mu \in \Omega^\ell(M)$.

6.9 Push-Forward

Definition 6.8. Let $f : M \to N$ be smooth. The push-forward (differential) of f at $p \in M$ is

$$f_* : \mathfrak{X}(M) \to \mathfrak{X}(N) : v \mapsto f_* v,$$

defined for $v \in \mathfrak{X}(M)$ and $g \in C^\infty(N)$ by

$$(f_* v)(g) \equiv v(g \circ f).$$

6.10 Pullback

Definition 6.9. Let $f : M \to N$ be smooth. The pullback map is the linear map

$$f^* : \Omega^k(N) \to \Omega^k(M) : \omega \mapsto f^* \omega,$$

defined for $\omega \in \Omega^k(N)$ and $v_1, \ldots, v_k \in \mathfrak{X}(M)$ by

$$(f^* \omega)(v_1, \ldots, v_k) \equiv \omega(f_* v_1, \ldots, f_* v_k).$$

Pullback map properties:

- Compatible with the wedge product: $f^*(\mu \wedge \omega) = f^*\mu \wedge f^*\omega$.
- Compatible with the exterior derivative: $f^*(d\omega) = d(f^*\omega)$.

Chapter 7

Integration of Differential Forms

7.1 Integration in $\mathbb{R}^3$

Definition 7.1. Let $\omega \in \Omega^1(D)$, let $C \subset D$ be smooth and orientable with parametrization $x(t) = \big(x(t), y(t), z(t)\big)$, $t \in [a, b]$. Then the integral of ω along C is

$$\int_C \omega \equiv \int_a^b \omega_{x(t)}(\dot{x}(t))\, dt.$$

It follows that by definition of ω,

$$\begin{aligned}\int_a^b \omega_{x(t)}(\dot{x}(t))\, dt &= \int_a^b \sum_{i=1}^3 F_i(x(t))\, dx^i(\dot{x}(t))\, dt = \int_a^b \sum_{i=1}^3 F_i(x(t))\dot{x}^i(t)\, dt \\ &= \int_a^b (F(x(t)), \dot{x}(t))\ dt = \int_C (F, \mathbf{t})\ ds.\end{aligned}$$

Definition 7.2. Let $\omega \in \Omega^2(D)$ and let $S \subset D$ be a smooth orientable surface with parametrization $x = x(u, v)$, $(u, v) \in G \subset \mathbb{R}^2$. Then the integral over S is defined by

$$\int_S \omega \equiv \iint_G \omega_{x(u,v)}\left(\frac{\partial x}{\partial u}, \frac{\partial x}{\partial v}\right)\ du\, dv.$$

It follows that

$$\begin{aligned}\iint_G \omega_{x(u,v)}\left(\frac{\partial x}{\partial u}, \frac{\partial x}{\partial v}\right)\ du\, dv &= \iint_G F_1(x(u,v))\, dy \wedge dz\left(\frac{\partial x}{\partial u}, \frac{\partial x}{\partial v}\right)\ du\, dv \\ &\quad + \iint_G F_2(x(u,v))\, dz \wedge dx\left(\frac{\partial x}{\partial u}, \frac{\partial x}{\partial v}\right)\ du\, dv \\ &\quad + \iint_G F_3(x(u,v))\, dx \wedge dy\left(\frac{\partial x}{\partial u}, \frac{\partial x}{\partial v}\right)\ du\, dv \\ &= \iint_G \left(F(x(u,v)), \frac{\partial x}{\partial u} \times \frac{\partial x}{\partial v}\right)\ du\, dv = \iint_S (F, n)\ d\sigma.\end{aligned}$$

7.1.1 1-form $\to$ curl (line $\to$ surface)

$$\omega = F_1\, dx + F_2\, dy + F_3\, dz.$$

$$d\omega = (\partial_y F_3 - \partial_z F_2)\, dy \wedge dz + (\partial_z F_1 - \partial_x F_3)\, dz \wedge dx + (\partial_x F_2 - \partial_y F_1)\, dx \wedge dy.$$

Identification:

$$d\omega \equiv (\operatorname{curl} F)_1\, dy \wedge dz + (\operatorname{curl} F)_2\, dz \wedge dx + (\operatorname{curl} F)_3\, dx \wedge dy.$$

Integral form:

$$\int_C \omega = \int_C F_i\, dx^i = \int_C (F, \mathbf{t})\ ds, \qquad \int_S d\omega = \iint_S (\operatorname{curl} F, n)\ d\sigma.$$

7.1.2 2-form → divergence (surface → volume)

$$\omega = F_1\, dy \wedge dz + F_2\, dz \wedge dx + F_3\, dx \wedge dy$$

implies

$$d\omega = (\partial_x F_1 + \partial_y F_2 + \partial_z F_3)\, dx \wedge dy \wedge dz.$$

Identification:

$$d\omega \equiv (\operatorname{div} F)\, dx \wedge dy \wedge dz.$$

Integral form:

$$\int_S \omega = \iint_S (F, n)\ d\sigma, \quad \int_D d\omega = \iiint_D \operatorname{div} F\, dV.$$

7.1.3 3-form (volume)

$$\omega = f\, dx \wedge dy \wedge dz \qquad \Rightarrow \qquad \int_D \omega = \iiint_D f\, dV.$$

7.2 Integration in $\mathbb{R}^n$

7.2.1 Parametrized k-Subsets

A set $M \subset D \subset \mathbb{R}^n$ is a parametrized k-subset if there exists

$$x : G \subset \mathbb{R}^k \to \mathbb{R}^n$$

such that x is injective and $M = x(G)$.

It is smooth if $x \in C^\infty$ and if the vectors

$$\frac{\partial x}{\partial u_1}, \frac{\partial x}{\partial u_2}, \ldots, \frac{\partial x}{\partial u_k}$$

are linearly independent.

7.2.2 Orientability

Let M be a smooth parametrized k-subset with parametrization

$$x : G \subset \mathbb{R}^k \to \mathbb{R}^n.$$

M is orientable if there exists a non-zero differential k-form β on M such that the parametrization is compatible:

$$\beta_x(T_{u_1}, T_{u_2}, \cdots, T_{u_k}) > 0 \qquad \forall (u_1, u_2, \cdots, u_k) \in G.$$

7.3 Integration on Manifolds

7.3.1 Integration of Top-Forms in a Single Chart

Let M be an oriented n-dimensional smooth manifold, and let $\omega \in \Omega^n(M)$ have compact support contained in a chart $(U, \varphi) \subset M$.

We define the integral of ω over M by pulling it back to $\mathbb{R}^n$:

$$\int_M \omega := \int_{\varphi(U)} (\varphi^{-1})^* \omega.$$

In coordinates, this becomes

$$(\varphi^{-1})^* \omega = f(x)\, dx^1 \wedge \cdots \wedge dx^n,$$

so that

$$\int_M \omega = \int_{\varphi(U)} f(x)\, dx^1 \cdots dx^n.$$

Note 7.3. The integral is well-defined since φ is a diffeomorphism and $\operatorname{supp}(\omega) \subset U$.

7.3.2 Independence of the Chart

Let (U, φ) and (V, ψ) be two charts such that $\operatorname{supp}(\omega) \subset U \cap V$.

Writing

$$h = \psi \circ \varphi^{-1},$$

we obtain, by change of variables,

$$\int_{\psi(V)} g(y)\, dy^1 \cdots dy^n = \int_{\varphi(U)} g(h(x)) \det Dh(x)\, dx^1 \cdots dx^n.$$

Thus the value of the integral does not depend on the chosen chart.

7.3.3 Partition of Unity

Let M be a smooth manifold. A collection of smooth functions $\{\chi_\alpha\}_{\alpha\in A} \subset C^\infty(M)$ is called a partition of unity if:

- $0 \le \chi_\alpha \le 1$,
- $\sum_{\alpha\in A} \chi_\alpha(p) = 1$ for all $p \in M$,
- the collection is locally finite.

The partition is subordinate to an open cover $\{U_\alpha\}$ if

$$\operatorname{supp}(\chi_\alpha) \subset U_\alpha.$$

Theorem 7.4. *Every open cover of a smooth manifold admits a subordinate smooth partition of unity.*

7.3.4 Bump Functions

A bump function is a smooth function with compact support that is identically equal to 1 on a smaller region.

Such functions are used to construct partitions of unity.

7.3.5 Integration Using Multiple Charts

Let $\omega \in \Omega^n(M)$ with

$$\operatorname{supp}(\omega) \subset \bigcup_{i=1}^{s} U_i,$$

where each U_i is a chart domain.

Let $\{\chi_i\}_{i=0}^s$ be a partition of unity subordinate to the open cover

$$\{U_0, U_1, \ldots, U_s\}, \qquad U_0 := M \setminus \operatorname{supp}(\omega).$$

Then

$$\int_M \omega = \int_M \left(\sum_{i=0}^{s} \chi_i\right) \omega = \sum_{i=0}^{s} \int_M \chi_i \omega.$$

Since $\chi_0\omega = 0$, this reduces to

$$\int_M \omega = \sum_{i=1}^{s} \int_M \chi_i \omega.$$

Each term satisfies

$$\int_M \chi_i \omega = \int_{\varphi_i(U_i)} (\varphi_i^{-1})^*(\chi_i \omega),$$

so the integral is computed locally in each chart.

Note 7.5. The result is independent of both the chosen charts and the partition of unity.

7.4 Regions with Boundary via Regular Values

Let M be a smooth manifold and $f \in C^\infty(M)$ such that 0 is a regular value.

Define the region

$$D := \{p \in M \mid f(p) \geq 0\},$$

its boundary

$$\partial D := \{p \in M \mid f(p) = 0\},$$

and its interior

$$D^\circ := \{p \in M \mid f(p) > 0\}.$$

Note 7.6. The boundary is given by a level set of f.

7.5 Generalized Stokes' Theorem

Let M be an oriented n-dimensional manifold and $D \subset M$ a region with boundary ∂D. Let $\omega \in \Omega^{n-1}(M)$ be such that $\operatorname{supp}(\omega) \cap D$ is compact.

Then

$$\int_D d\omega = \int_{\partial D} \omega.$$

The boundary integral is interpreted as

$$\int_{\partial D} \omega = \int_{\partial D} \iota^* \omega,$$

where $\iota : \partial D \hookrightarrow M$ is the inclusion map.

Intuition 7.7. The exterior derivative $d\omega$ measures the infinitesimal behaviour of ω inside D, while the integral over ∂D captures its global effect on the boundary.